

# Xinghao Chen

Ph.D. Student in Computer Science

## Research Interests:

Long-Context KV Cache Reuse

Latent-Space Reasoning

Thought Compression & Distillation



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🏠 [Google Scholar](#)

## EDUCATION

**The Hong Kong Polytechnic University & Eastern Institute of Technology, Ningbo**

*Joint Ph.D. Student in Computer Science*

Hong Kong / Ningbo, China

Sep. 2024 – Present

**Supervisors:** [Wenjie Li](#), Professor   [Xiaoyu Shen](#), Assistant Professor

**Nankai University, College of Artificial Intelligence**

*B.S. in Intelligent Science and Technology*

Tianjin, China

Sep. 2019 – Jun. 2023

## INTERNSHIP EXPERIENCE

**Tencent Youtu Lab**

*Research Intern   Working with: [Junnan Dong](#)*

Jun. 2026 – Present

## PUBLICATIONS

\* indicates equal contribution, † indicates corresponding author.

### Preprints

- [1] [X. Chen\\*](#), J. Dong\*, C. Ke, C.T. Leong, H. Sun, K. Chen, S. An, R. Qiao, X. Sun, W. Li, X. Shen†. [Attuner: Recomputation-Free KV Cache Reuse via Query-Side Adaptation](#). **Preprint**.
- [2] C. Ke, [X. Chen](#), X. Shen, K. Chen, S. An, J. Dong, R. Xu, R. Qiao, X. Sun. [Disentangling Long-Term Memory via Latent Neuro-Symbolic Reasoning](#). **Preprint**.
- [3] C. Ke, [X. Chen](#), Y. Bei, Y. Lai, X. Shen, K. Chen, R. Qiao, S. An, J. Dong, R. Xu, X. Sun. Latent Multimodal Memory Reasoning. **Preprint**.

### LLM Reasoning & Efficient Intelligence

- [4] [X. Chen](#), C.T. Leong, W. Guo, J. Wang, W. Li, X. Shen†. What Makes Effective Supervision in Latent Chain-of-Thought: An Information-Theoretic Analysis. [C]// (ICML) Forty-Third International Conference on Machine Learning. **CCF-A**  
Established an information-theoretic framework dissecting latent CoT supervision into Trajectory Control and State Alignment; introduced the Unified Latent Probe (ULP) to reveal a strict information-performance binding between mutual information retention and reasoning accuracy.
- [5] [X. Chen\\*](#), A. Zhao\*, H. Xia, X. Lu, H. Wang, Y. Chen, W. Zhang, J. Wang†, W. Li, X. Shen†. [Reasoning Beyond Language: A Comprehensive Survey on Latent Chain-of-Thought Reasoning](#). [C]// (EMNLP) Findings of the 2026 Conference on Empirical Methods in Natural Language Processing. **CCF-B**  
The first comprehensive survey on latent CoT reasoning, categorizing methods along two axes: *Token-wise Horizontal Level* and *Layer-wise Vertical Level*; maintaining a continuously updated repository at [Awesome-Latent-CoT](#).
- [6] [X. Chen](#), Z. Sun, W. Guo, M. Zhang, Y. Chen, Y. Sun, H. Su, Y. Pan, D. Klakow, W. Li†, X. Shen†. [Unveiling the Key Factors for Distilling Chain-of-Thought Reasoning](#). [C]// (ACL) Findings of the 2025 Annual Meeting of the Association for Computational Linguistics. **CCF-A**  
Conducted systematic study on *what makes effective Chain-of-Thought supervision for smaller models*, revealing that the conventional wisdom of “stronger teacher, better student” does not hold in reasoning distillation.
- [7] S. Lyu\*, [X. Chen\\*](#), Z. Sun, T. Liu, D. Zhu, X. Shen. [When Compression Helps and When It Hurts: Condition-Aware Analysis of Chain-of-Thought Distillation](#). [C]// (EMNLP) The 2026 Conference on Empirical Methods in Natural Language Processing. **CCF-B**
- [8] H. Chen\*, J. Lin\*, [X. Chen](#), Y. Fan, X. Jin, H. Su, J. Dong, J. Fu, X. Shen. [Rethinking Visual Layer Selection in Multimodal LLMs](#). [C]// (EMNLP) The 2025 Conference on Empirical Methods in Natural Language Processing. **CCF-B, Oral**
- [9] Y. Chen, D. Zhu, Y. Sun, [X. Chen](#), W. Zhang, X. Shen. [The Accuracy Paradox in RLHF: When Better Reward Models Don't Yield Better Language Models](#). [C]// (EMNLP) The 2024 Conference on Empirical Methods in Natural Language Processing. **CCF-B**
- [10] X. Lu, S. Liu, B. Yin, Y. Li, [X. Chen](#), H. Su, Y. Jin, W. Zeng, X. Shen. [MultiConIR: Towards Multi-Condition Information Retrieval](#). [C]// (EMNLP) Findings of the 2025 Conference on Empirical Methods in Natural Language Processing. **CCF-B**
- [11] H. Wu, X. Wang, J. Zhang, J. Tong, [X. Chen](#), J. Lin, Y. Ma, X. Shen. [UTPTrack: Towards Simple and Unified Token Pruning for Visual Tracking](#). [C]// (CVPR) The 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition. **CCF-A**
- [12] J. Yang, [X. Chen](#), Y. Liu, Z. Sun, W. Guo, X. Shen. [AutoDataset: A Lightweight System for Continuous Dataset Discovery and Search](#). Arxiv, 2603.07271.

### Applied AI (Pre-doctoral)

- [13] X. Zhang, [X. Chen](#), Y. Ding, Y. Zhang, Z. Wang, J. Shi, N. Johansson, X. Huang. [Smart Real-Time Evaluation of Tunnel Fire Risk and Evacuation Safety via Computer Vision](#). *Safety Science*, 2024. **CAS Q1, JCR Q1**

- [14] Y. Ding, X. Chen, Y. Zhang, X. Huang. [Smart Building Evacuation by Tracking Multi-Camera Network and Explainable Re-Identification Model](#). *Engineering Applications of Artificial Intelligence*, 2025. **CAS Q1, JCR Q1**
- [15] Y. Ding, X. Chen, Z. Wang, Y. Zhang, X. Huang. [Human Behaviour Detection Dataset \(HBDset\) Using Computer Vision for Evacuation Safety and Emergency Management](#). *Journal of Safety Science and Resilience*, 2024.
- [16] X. Chen, G. Cheng, S. Liu, S. Meng, Y. Jiao, W. Zhang, J. Liang, W. Zhang. Probing 1D Convolutional Neural Network Adapted to Near-Infrared Spectroscopy for Efficient Classification of Mixed Fish. *Spectrochimica Acta Part A*, 2022. **CAS Q2, JCR Q1**

## RESEARCH PROJECTS

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### Position-Independent KV Cache Reuse for Long-Context Agents

Jun. 2026 – Present

*Ongoing research*

Developed *Attuner* [1] for position-independent caching (PIC) of reusable skills, documents, and memories. Identified attention routing as a source of quality loss and trained low-rank query adapters through full-context distillation to read frozen artifact KV states without recomputation. Recovers near-full-prefill quality at near-PIC warm-cache time to first token (TTFT).

#### Future directions:

- **Quality:** Validate PIC with repeated artifact injection across multi-turn dialogues; address artifact discovery and selection under distracting context after improving evidence utilization.
- **Efficiency:** Measure and optimize cache construction, storage I/O, transfer, and assembly costs; characterize the quality–latency–memory Pareto frontier across cache budgets and reuse frequencies.
- **Goal:** Enable reusable agent context that matches full-prefill quality at PIC-level latency within practical storage and bandwidth budgets.

### Latent Memory Reasoning for Long-Term Multimodal Agents

Feb. 2026 – Present

*Ongoing research*

Contributed to neuro-symbolic approaches that organize long-term interaction histories in latent space. LGM [2] disentangles textual memories into sparse concepts and builds query-conditioned graphs to reason over explicit and implicit preferences. LMMR [3] aligns text states and visual patches in a shared sparse concept space, models cross-modal and temporal relations in dialogue-turn graphs, and injects graph readouts into continuous latent reasoning while retaining native visual tokens for fine-grained, temporally grounded responses.

### Latent Chain-of-Thought Reasoning

Jan. 2025 – Feb. 2026

*Core research direction during Ph.D.*

Investigating how to internalize explicit reasoning into continuous latent spaces for efficient and faithful LLM inference. Established an information-theoretic framework for analyzing supervision signals in latent CoT [4]; authored the first comprehensive survey of the field [5]; and empirically identified key factors governing CoT distillation from large to small models [6]. Ongoing work explores extending latent reasoning representations to serve as compressed, reusable memory for agentic systems.

## SERVICE

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### Teaching Assistant, The Hong Kong Polytechnic University

- COMP 3122: Information Systems Development Spring 2026
- COMP 5241: Software Engineering and Development Fall 2025
- COMP 1411: Introduction to Computer Systems Spring 2025
- COMP 5511: Artificial Intelligence Concepts Fall 2024

## INVITED TALKS

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### Reasoning Beyond Language: A Comprehensive Survey on Latent CoT Reasoning

*Invited talk at [NICE](#) and [CIPS](#) [Video]*

*Jun. 2025*

## SKILLS

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- **Programming Languages:** Python, MATLAB, C++
- **LLM Frameworks:** LLaMA-Factory, VERL, vLLM, SGLang
- **AI-Assisted Development:** Claude Code, Cursor, OpenAI Codex, OpenClaw
- **Languages:** Mandarin (Native), English (Fluent)

## SELF-EVALUATION

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- Endlessly curious about new ideas and technologies. Committed to research that matters beyond the paper — bridging the gap from insight to real-world impact. Collaborative, outgoing, and motivated by a simple standard: work that is technically sound, practically grounded, and genuinely worth sharing.